INCH-POUND

MIL-DTL-43002E
14 September, 2000

SUPERSEDING
MIL-S-43002D
6 February, 1984

DETAILED SPECIFICATION

SHIELD, STOVEPIPE, TENT, NONMETALLIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for a flexible, silicone rubber coated shield for a tent stovepipe opening.

2. APPLICABLE DOCUMENTS

*2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

*2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions, clarifications) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center Philadelphia, Clothing and Textiles Directorate, Attn: DSCP-CNR, 700 Robbins Ave., Philadelphia, PA 19111-5096 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8340

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SPECIFICATIONS

FEDERAL

A-A-55280 - Cloth, Coated, Glass, Silicone Rubber

(Unless otherwise indicated. copies of federal and military specifications, standards, and handbooks are available from the Document Automation and Production Service, (DAPS) Bldg. 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

U.S. ARMY BIOLOGICAL & CHEMICAL COMMAND

8-1-10 - Shield, Stovepipe, Tent, Nonmetallic; Assembly and Details

(Copies of drawings are available from the U. S. Army Biological Chemical Command, ATTN: AMSSB-REO-S(N), NATICK, MA 01760-5017.)

*2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-412	- Tensile Strength
ASTM-D-573	- Air Heat Test, Air Heating Medium
ASTM-D-1053	- Stiffness, Torsional, Low-Temperature
ASTM-D-2240	- Hardness Durometer
ASTM-D-2256	- Tensile Properties of Yarns by the Single-Strand Method

(Applications for copies of referenced documents should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19426-2959.)

*2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

*3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.2).

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3.2 Materials and components. The materials (see 6.5) and components shall be as specified on the applicable drawings and as specified herein.

*3.2.1 Coated cloth. The coated cloth for making the main piece of the shield shall be silicone rubber coated glass cloth conforming to A-A-55280.

*3.2.2 Rubber. The rubber for the molded collar shall be compounded to conform to the requirements specified in Table I and shall be pigmented such that the color approximates Camouflage Green 483.

Table I. Physical requirements for rubber of molded collar

Characteristic	Requirement
Tensile strength lbs/in^2 (min)	
Before aging	600
After aging	400
Ultimate elongation % (min)	
Before aging	175
After aging	125
Hardness, scale reading:	
Before aging	60 ± 5
After aging, maximum increase, %	25
Cold resistance at -85°F, minimum angular twist, °	1/

1/ The angular twist of the specimens in degrees shall be not less than that specified for the specimen thickness in ASTM D 1053.

3.3 Construction. At the option of the contractor, the construction of the shield shall be in accordance with construction A or B as specified in 3.3.1 and 3.3.2.

3.3.1 Construction A. The shield shall consist of two parts: the main piece and the molded collar. The dimensions of the shield shall be as specified in Drawing 8-1-10 and the cross-sectional configuration shall be as specified for construction A in Drawing 8-1-10. The main piece shall be made of the coated cloth specified in 3.2.1. The collar shall be made of the rubber specified in 3.2.2. The hole edge of the main piece shall be inserted into the rubber collar. The collar shall then be stitched to the main piece with one continuous row of stitching, 1/8 to 3/16 inch from the outside edge of the collar. Stitching shall be stitch type 301, 4 to 5 stitches per inch. The thread shall be as specified in 3.3.1.1. The last end of the stitching shall be run over the first end for a distance of 1 to 1-1/2 inches. Thread ends shall be trimmed to not more than 1/8 inch.

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3.3.1.1 Thread. The thread for stitching the collar to the main piece shall be polytetraflouroethylene (TFE) coated glass thread, natural color equal to or better than Dodge Industries Fluoroglass Thread M753-18. The thread shall meet the following requirements (see 4.4.1):

Breaking strength, lbs., (min)	15
Length/pound, yds.	
(min)	1800
(max)	2400
TFE content, % by weight (min)	12

3.3.2 Construction B. The shield shall consist of two parts: the main piece and the molded collar. The dimensions of the shield shall be as specified in Drawing 8-1-10 and the cross-sectional configuration shall be as specified for construction B in Drawing 8-1-10. The main piece shall be made of the coated cloth specified in 3.2.1. The collar shall be made of the rubber specified in 3.2.2. The collar shall be molded by the means of vulcanizing over the main piece and there shall be no visible evidence of cloth showing on the inside edge of the molded collar.

3.4 Strength of collar. There shall be no separation of the collar from the main piece or tearing away of the rubber in the collar when tested as specified in 4.5.1.

*3.5 Identification label. Each shield shall be legibly marked along the edge on the back or reverse side with black ink in letters not less than ½ inch in height with the following information:

Stock Number
Nomenclature
Contract Number
Date of Manufacture

*3.6 Replacement of defective components. During the manufacturing process, components having material defects or damages that are classified as defects in 4.4.3 and 4.4.4 shall be removed from production and replaced with non-defective or properly matched components.

*3.7 Workmanship. The finished shields shall conform to the quality of product established by this document.

4. VERIFICATION

4.1 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all dimensions specified have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

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4.1.2 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2) it shall be examined for the defects listed in 4.4.3 and 4.4.4 and tested as specified in 4.4.5. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Conformance inspection. Sampling for inspection shall be as specified in the contract or purchase order.

*4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document. A certificate of compliance shall be furnished with each shipment or lot stating that the materials and components conform to the applicable requirements listed below:

Material component	Requirement
Thread (composition, breaking strength) 1/ length/lb., TFE content	3.3.1.1

1/ *When tested in accordance with ASTM-D-2256. Testing speed shall be 12 ± 0.5 in/min and a 10 inch gauge length shall be used.

4.4.2 Rubber test. In addition to the requirements of the applicable referenced documents the rubber shall be tested for the characteristics listed in Table II. The lot size shall be expressed in units of one shield. The sample unit shall be four pieces of the rubber compound for the collar, molded and vulcanized under the same conditions as the finished shields. Each piece shall be 6 by 6 inches by 0.075 ± 0.010 inch thick. The sample size shall be as specified below. The lot shall be unacceptable if any sample unit fails to meet any requirement specified. A certificate of compliance shall be furnished with each shipment or lot stating that the test pieces of the rubber compound were molded and vulcanized under the same conditions as the finished shield.

Lot size (shields)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table II. Rubber tests

Characteristic	Requirement paragraph	Test method
Tensile strength:		
before aging	Table I	ASTM-D-412 1/
after aging	Table I	ASTM-D-412 1/ & ASTM-D-573 2/

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Table II. Rubber tests

Characteristic	Requirement paragraph	Test method
Ultimate elongation:		
before aging	Table I	ASTM-D-412 1/
after aging	Table I	ASTM-D-412 1 & ASTM-D-573 2/
Hardness		
before aging	Table I	ASTM-D-2240 3/
after aging	Table I	ASTM-D-2240 3/ & ASTM-D-573 2/
Cold resistance	Table I	ASTM-D-1053 4/

1/ Die No. III shall be used to cut test specimens.
2/ Except that test specimens shall be aged for 3 hours at 600°F.
3/ The Shore-A-Durometer shall be used to measure the hardness.
4/ The temperature of the exposure shall be -85 °F and the heat transfer medium shall be methanol. The exposure time shall be 20 minutes.

4.4.3 End item visual examination. The end item shall be examined for the defects listed in Table III and applicable to the construction (A or B) furnished. The lot size shall be expressed in units of shields. The sample unit shall be one shield. The inspection level shall be II.

TABLE III. End item visual defects

Examine	Defect	Classification Major	Minor
Main piece	Tear, cut or hole (excepting pinhole)	101	
	Abrasion, resulting in thin or weak place or exposure of uncoated cloth structure	102	
	Blister or soft lump	103	
	Hard lump exceeding ½ inch in combined length and width directions, clearly noticeable		201
	Crease or wrinkle resulting in doubling that cannot be corrected by manual pressure or adhesion of surfaces against each other	104	
	Area devoid of coating, measuring more than 1 inch in any lineal direction	105	
	Area devoid of coating, measuring 1 inch or less in any lineal direction		202
	Color not as specified		203
	Coating tacky	106	
Rubber collar	Crack, check, pit or blister	107	
	Tear, void, sunburst, lump or shrinkage mark	108	
	Color of molded rubber collar does not approximate camouflage green 483		204
	Shade does not match approved guide sample		205
Thread	Not natural color.		206

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TABLE III. End item visual defects (continued)

Examine	Defect	Classification Major	Minor
Design	Any characteristic not in accordance with specified requirements (unless otherwise indicated herein)	109	
Construction & Workmanship	Any component missing	110	
	Edge of cloth visible through inside edge of molded collar	111	
	Wrong stitch type	112	
	Stitches per inch less than 4 or more than 5	113	
	Stitching margin less than 1/8 or more than ¼ inch		207
	Overstitching of first end:		
	- less than 1 inch but not less than ½ inch or more than 1 ½ inches		208
	- less than ½ inch	114	
	Thread ends not trimmed to within 1/8 inch		209
	Broken and skipped stitches:		
	- more than one consecutively or more than three per shield	115	
	- more than one but not more than three provided the broken stitches are at least 3 inches apart		210
	Run off	116	
	Tight stitching showing obvious splitting of rubber collar	117	
	Loose stitching, thread loops on top or bottom surfaces which can be entered by a size 18 needle	118	
	Delamination or separation of collar from main piece:		
	- greater than 1 inch	119	
	- 1 inch or less		211
	Appreciable amount of rubber from molded collar distributed on sides of reinforcing rings seriously affecting appearance		212
	Malformation, irregularity or poor trim of molded collar	120	
	Spot, stain, streak or other discoloration clearly visible		213
	Overall uncleanness		214
	Objectionable odor		215
	NOTE: Aromatic odors usually associated with Specific types of rubber compounds shall not be regarded as objectionable		
Marking	Smeared, missing, incomplete, incorrect or illegible	121	
	Not in proper location		216
	Height of lettering not as specified		217

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*4.4.4 End item dimensional examination. The end item shall be examined for conformance to the specified dimensions. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of shields. The sample unit shall be one shield. The inspection level shall be S-3.

4.4.5 End item testing. The end item shall be tested as specified in 4.5.1 for conformance to the collar strength requirements specified in 3.4 The lot size shall be expressed in units of shields. The sample size shall be the number of shields indicated by inspection level S-2. Any test failure shall be cause for rejection of the lot.

4.4.6 Packaging examination. The fully packaged end items shall be examined for marking, materials, workmanship and content defects.

4.5 Methods of inspection.

4.5.1 Strength of collar test. The strength of the collar shall be determined as follows:

Specimen – One shield. Two determinations shall be made on each shield.

Apparatus – Tensile testing machine specified in ASTM D 412 except as follows:

A hand vise clamp with jaws 1 ½ ± 1/16 inches wide (Millers Falls No 3 or equivalent) shall be attached to the upper grip. The lower grip shall be a clamp with jaws measuring 5 ½ inches ± 1/8 inch by 1 ¼ inches ± 1-1/8 inches. The rate of travel for the power actuated grip shall be 2.0 inches + 0.1 inch/minute.

Construction A. Insert the main piece of the shield between the upper gripping surfaces so that the edge of the collar is just above the gripping surfaces and the edge of the shield closest to the gripped part is perpendicular to the direction of travel. Clamp that edge of this shield in the lower grip and adjust the upper grip opening to 3/64 ± 1/64 inch. (see illustration for testing on drawing 8-1-10 typical for A or B construction). The second determination shall be made with the shield in a position 90^{o} to that of the first.

Construction B. Insert the thin portion of the rubber collar between the upper gripping surfaces so that the thick portion of the collar is just above the gripping surfaces and the edge of the shield closest to the gripped part of the collar is perpendicular to the direction of travel. Clamp that edge of the shield in the lower grip and adjust the upper grip opening to 3/16 ± 1/64 inch. (see illustration for testing on drawing 8-1-10 typical for A or B construction). The second determination shall be made with the shield in a position 90^{o} to that of the first.

Report - The specimen shall be considered as failing with respect to bond strength if failure occurs in either of the two determinations. Test failure is defined as separation of the collar from the main piece or tearing away of the rubber in the collar to the extent that application of 100 pounds of force cannot be attained. When a result of less than 100 pounds is caused by other than the two stated conditions, that determination will be considered invalid and an additional determination shall be performed.

5. PACKAGING

*5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory

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Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

*(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The stovepipe shields are designed for installation in a tent roof through which a stovepipe passes. The shield acts as insulation to protect the tent and canvas against excessive heat.

*6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When first article is required (see 3.1, 4.3 and 6.4).
d. Levels of preservation and packing (see 5.1).

6.3 Sample. For access to standard sample, address the contracting activity issuing the invitation for bids.

6.4 Rubber composition. A silicone rubber composition has been found to satisfactorily meet the requirements of Table I (see 3.2.2).

6.5 First article. When a first article is required, it will be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

*6.6 Equal item. Prior to the use of an “or equal” item, the supplier will submit the item, with supporting data to the contracting officer for subsequent approval or disapproval by the responsible military agency.

*6.7 Subject term (key word) listing.
Cover, protective
Rubber, silicone coated

*6.8 Changes from previous issue. The margins of this document have been marked with an asterisk(*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:
Army - GL
Air Force – 99

Review activities:
Army – MD
Air Force – 11.6, 82

Preparing activity
DLA-CT
Project No: 8340-0598

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7 and send to preparing activity.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-43002E	**2. DOCUMENT DATE** ***(YYYYMMDD)*** 2000/09/14

3. DOCUMENT TITLE

SHIELD, STOVEPIPE, TENT, NONMETALLIC

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(If applicable)*	7. DATE SUBMITTED *(YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME DEFENSE SUPPLY CENTER PHILADELPHIA DSCP-CNRP	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN (215) 737-3290 444-3290
c. ADDRESS *(Include Zip Code)* 700 Robbins Ave (Bldg 6, C&T) PHILADELPHIA, PA 19111-5092	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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